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Predictive operation of a hydrogen refuelling station with scheduled clients

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Outline

1 **Context**
Hydrogen refuelling stations · the research question

2 **Proposal**
A supervisory hybrid MPC · lexicographic objectives

3 **Availability**
Cascaded refuelling · the usual premise · patterns · the objective

4 **Bookings**
How a booking enters the model, tank by tank

5 **Results**
A preliminary six-hour case, three booked clients

6 **Conclusions**
What this paper answers · what remains

Hydrogen Refuelling Stations (HRSs)

Main characteristics: flexible in sourcing, complex in preparation

1

FLEXIBILITY

Hydrogen sourcing

on site

off site

2

FLEXIBILITY

Electrical energy sourcing

renewables

storage

grid

market

3

COMPLEXITY

Pressurised storage and compression

several pressure levels

compressors

4

COMPLEXITY

Cascaded refuelling

tanks in sequence

filling and refuelling

Hydrogen Refuelling Stations (HRSs)

Hydrogen sourcing

1 Hydrogen sourcing
produced on site, or delivered



Which source, and when, is a decision.

2
FLEXIBILITY
Electrical energy sourcing

3
COMPLEXITY
Pressurised storage and compression

4
COMPLEXITY
Cascaded refuelling

Hydrogen Refuelling Stations (HRSs)

Electrical energy sourcing

1

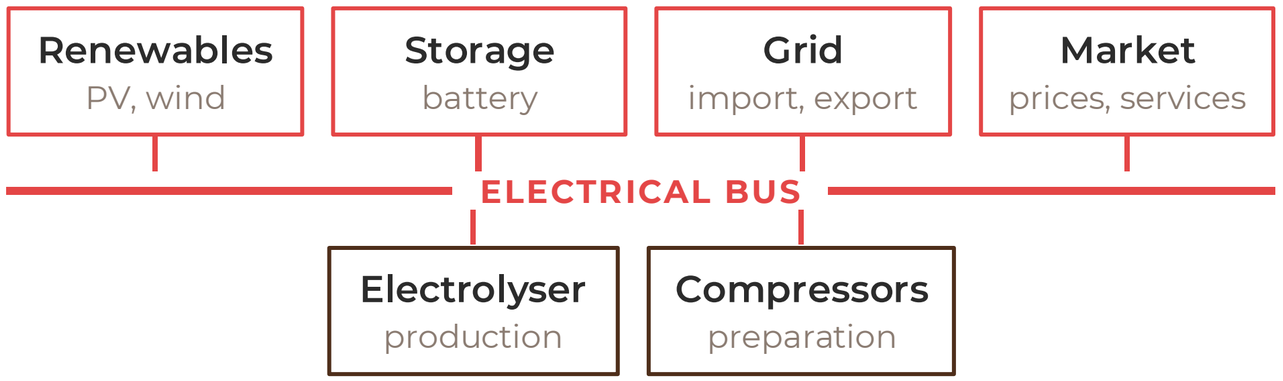
FLEXIBILITY

Hydrogen sourcing

2

Electrical energy sourcing

one bus, several sources, two loads



Production and compression are the loads.

3

COMPLEXITY

Pressurised storage and compression

4

COMPLEXITY

Cascaded refuelling

Hydrogen Refuelling Stations (HRSs)

Pressurised storage and compression

1

FLEXIBILITY

Hydrogen sourcing

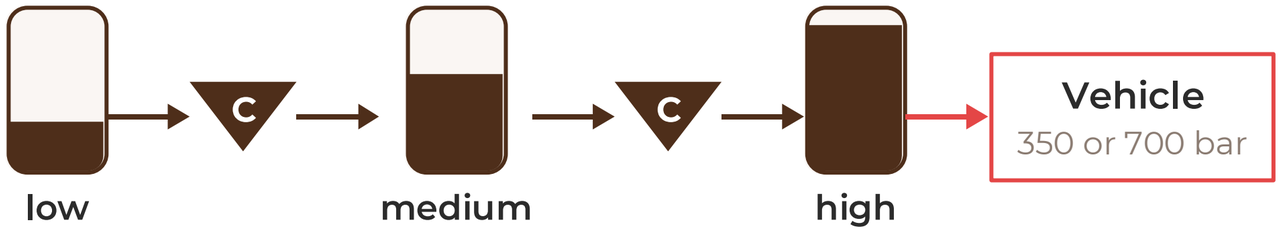
2

FLEXIBILITY

Electrical energy sourcing

3 **Pressurised storage and compression**

lifted in stages to the vehicle standard



Which tank to fill, and when, is a decision.

4

COMPLEXITY

Cascaded refuelling

Hydrogen Refuelling Stations (HRSs)

Cascaded dispensing

1
FLEXIBILITY
Hydrogen sourcing

2
FLEXIBILITY
Electrical energy sourcing

3
COMPLEXITY
Pressurised storage and compression

4 **Cascaded refuelling**
tanks opened in order, no compressor



Order matters, in filling and in refuelling.

Hydrogen Refuelling Stations (HRSs)

The product needs preparation

1

FLEXIBILITY

Hydrogen sourcing

on site

off site

2

FLEXIBILITY

Electrical energy sourcing

renewables

storage

grid

market

3

COMPLEXITY

Pressurised storage and compression

several pressure levels

compressors

4

COMPLEXITY

Cascaded refuelling

tanks in sequence

filling and refuelling

Source: which, when?

Compress: which tank, when?

Hydrogen needs preparation before delivery: pressure, order, timing.

RESEARCH QUESTION

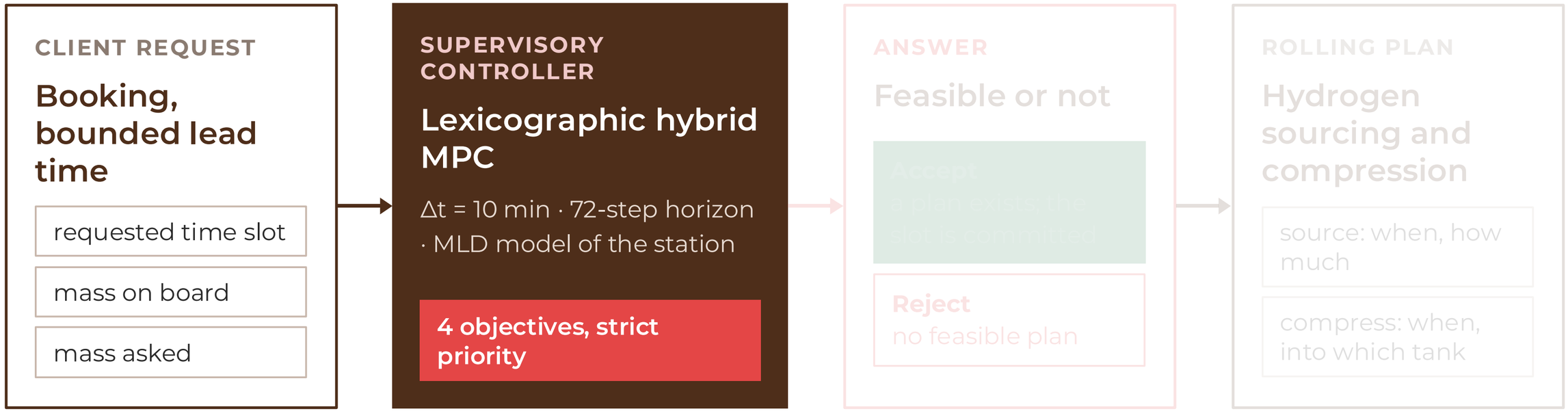
Can integrating client demand into the operation of the station bring quantifiable gains in energy efficiency and demand satisfaction?

FIRST STEP

Can the controller commit to a booked client in advance and still plan freely?

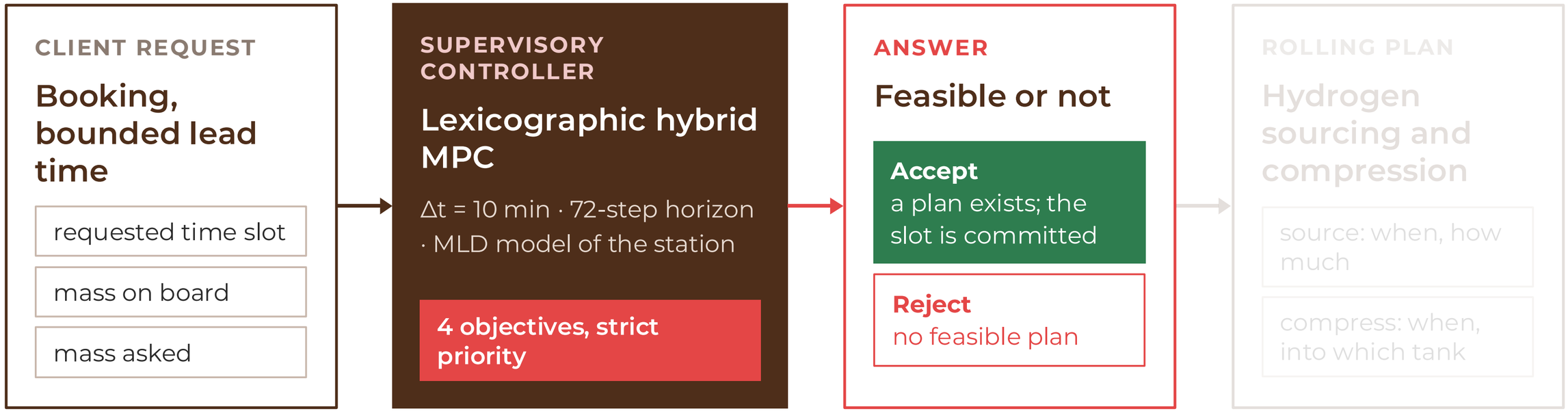
The proposed approach

A request arrives



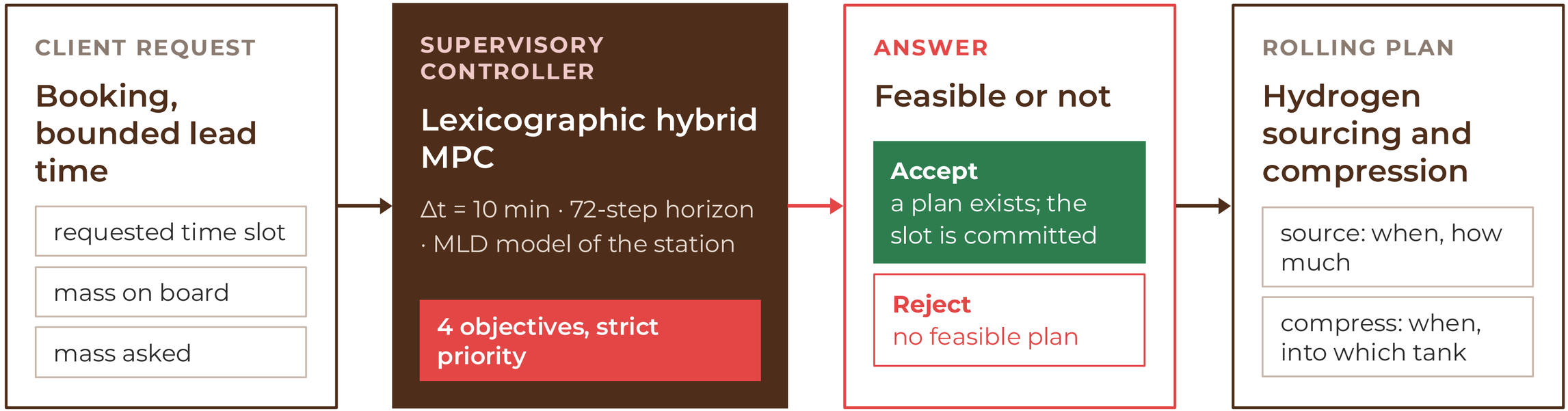
The proposed approach

Feasible or not



The proposed approach

A rolling plan



Lexicographic Control Objectives

Four objectives, strict priority

1 **Hydrogen availability**
Ready for unannounced arrivals

2 **Grid import**
Compression follows the PV

3 **Off-site H_2 purchase**
Trailer deliveries

4 **Surplus H_2 sale**
Monetise last

Lexicographic: strict priority, no weights

Service to booked clients: hard constraint

Bookings are guaranteed by the model, tank by tank. Grid import leads; availability keeps idle hours open to short-lead requests.

Lexicographic Control Objectives

Bookings enter as constraints

1 **Hydrogen availability**
Ready for unannounced arrivals

2 **Grid import**
Compression follows the PV

3 **Off-site H_2 purchase**
Trailer deliveries

4 **Surplus H_2 sale**
Monetise last

Lexicographic: strict priority, no weights

Service to booked clients: hard constraint

Bookings are guaranteed by the model, tank by tank. Grid import leads; availability keeps idle hours open to short-lead requests.

Lexicographic Control Objectives

Grid first, availability second

1 **Grid import**
Compression follows the PV

2 **Hydrogen availability**
Ready for unannounced arrivals

3 **Off-site H_2 purchase**
Trailer deliveries

4 **Surplus H_2 sale**
Monetise last

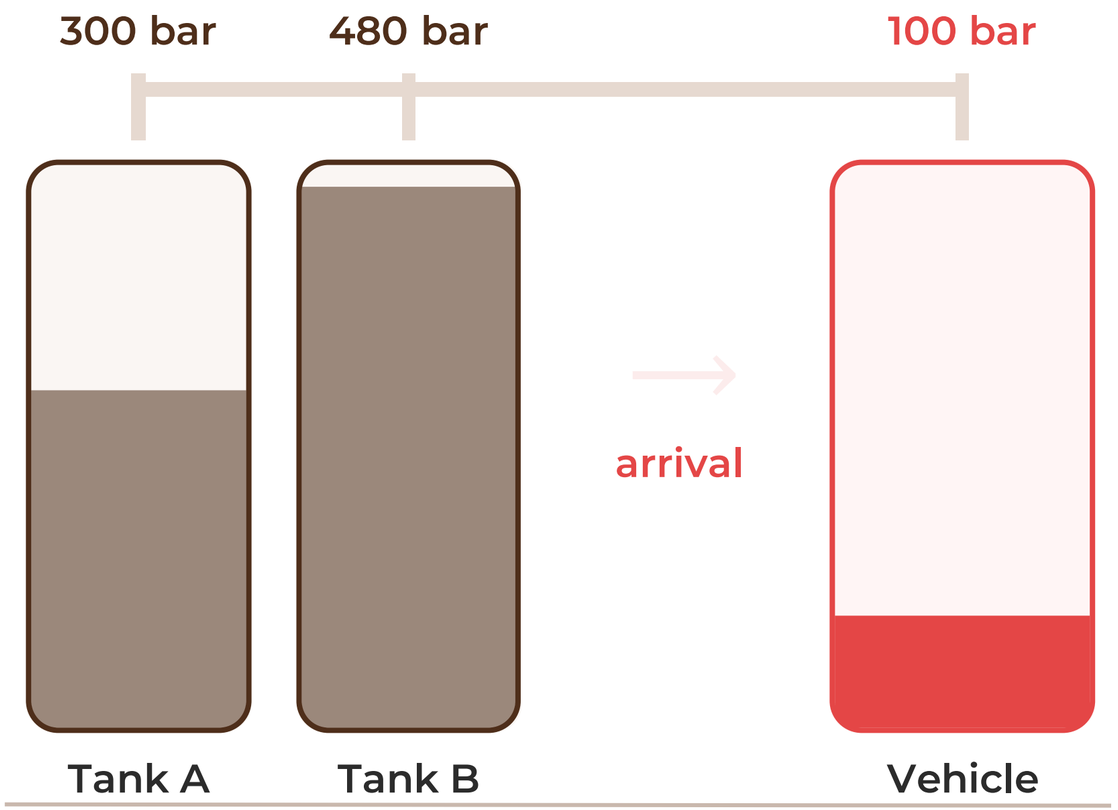
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Cascaded refuelling

Flow follows the pressure gradient

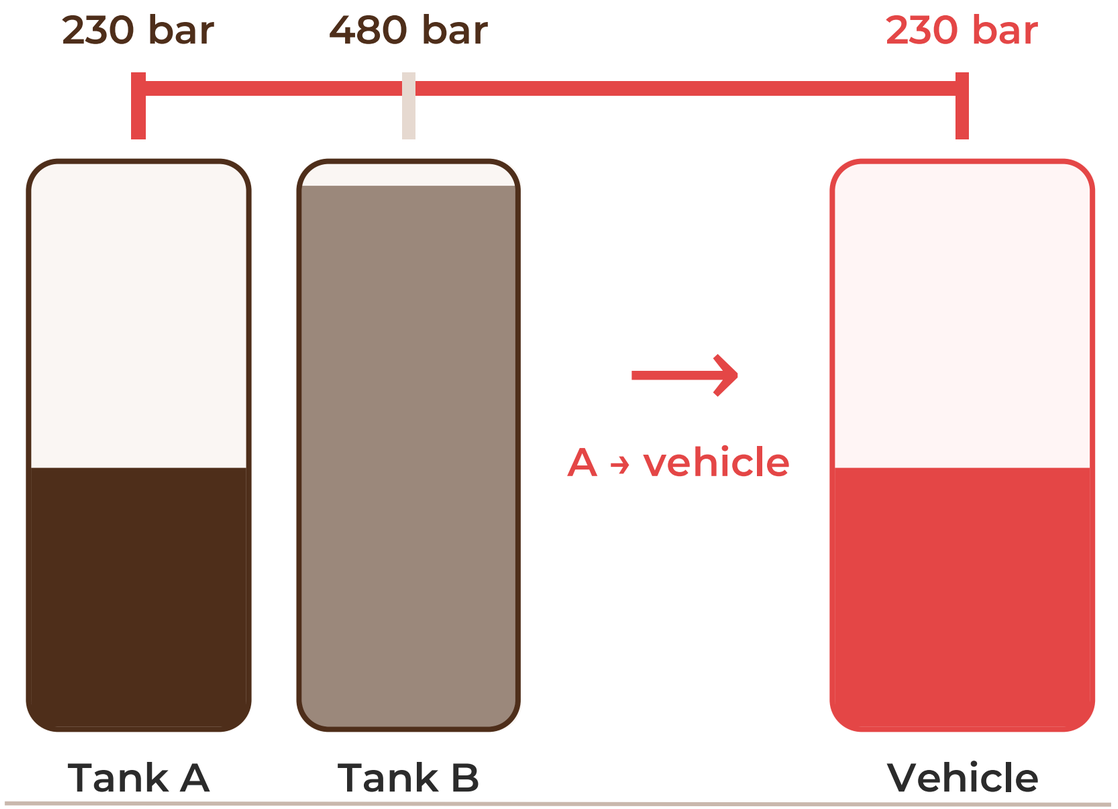


1. Open A: flow to $P_{eq,1}$, A gives Δm_A
2. No gradient: A cannot push more, whatever it still holds
3. Open B: flow to 350 bar, B delivers Δm_B
4. Order matters: **B → A** stalls at B's equilibrium

Delivered mass = $\Delta m_A + \Delta m_B$, set by pressures and volumes, not by stored mass.

Cascaded refuelling

Flow follows the pressure gradient

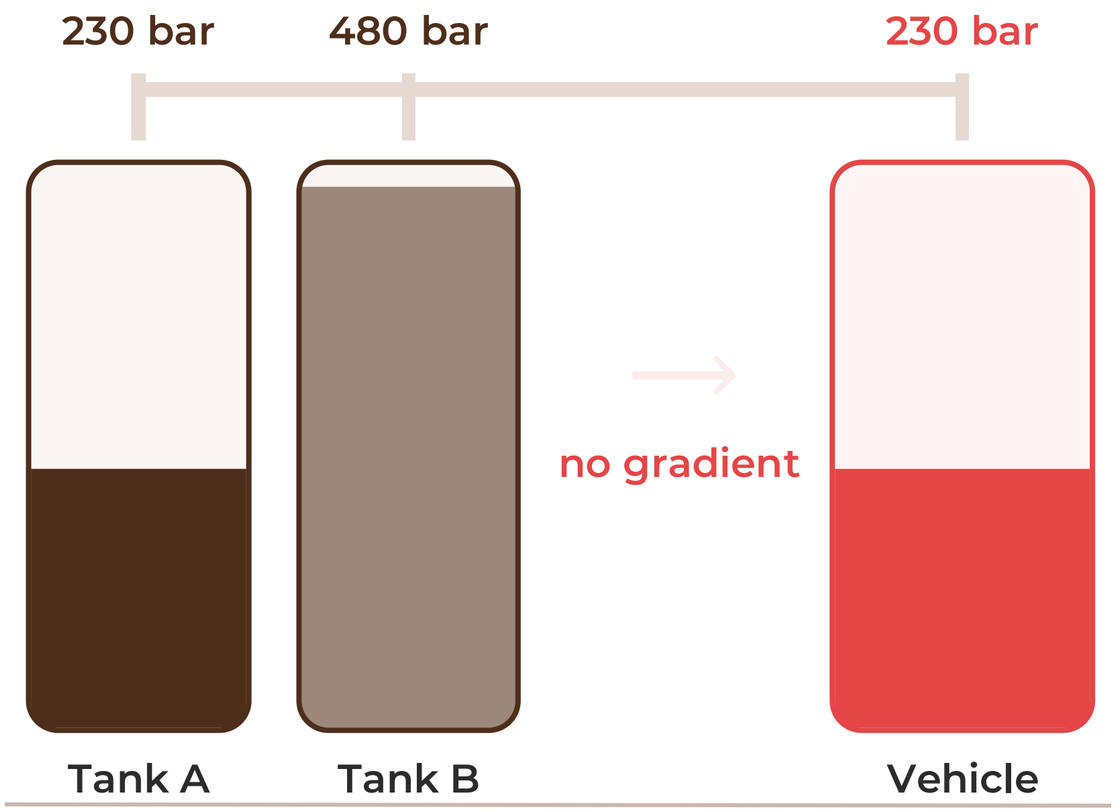


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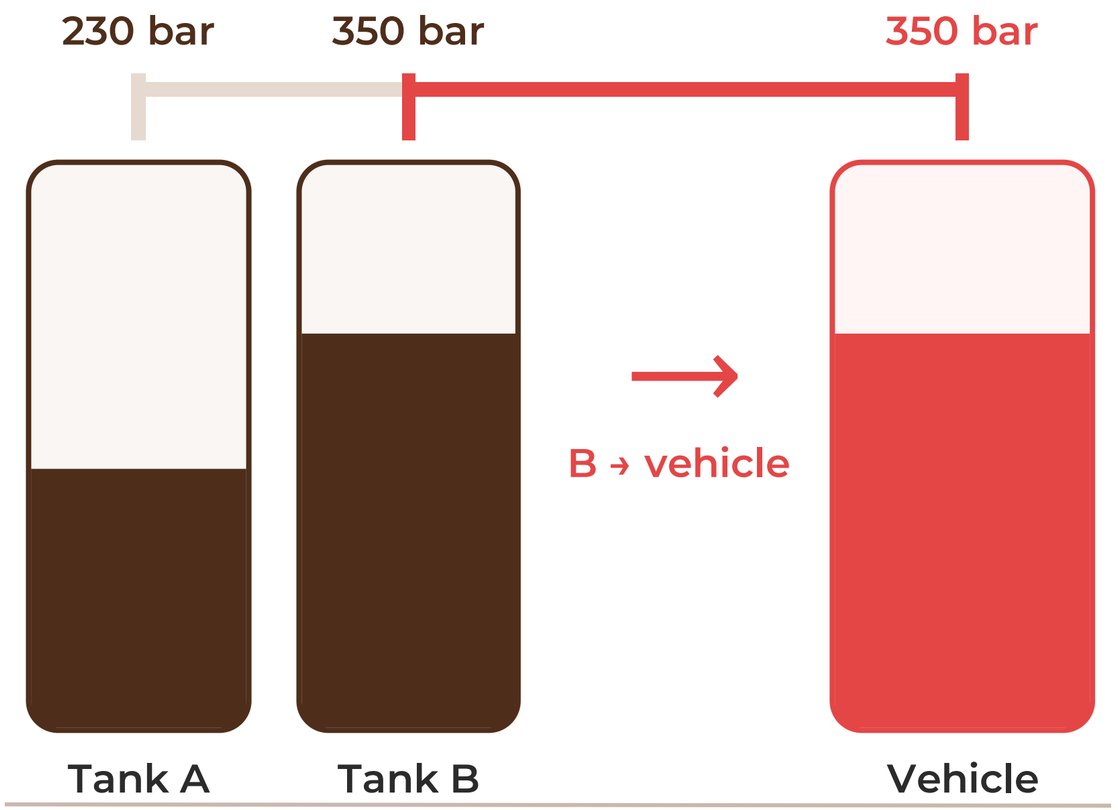


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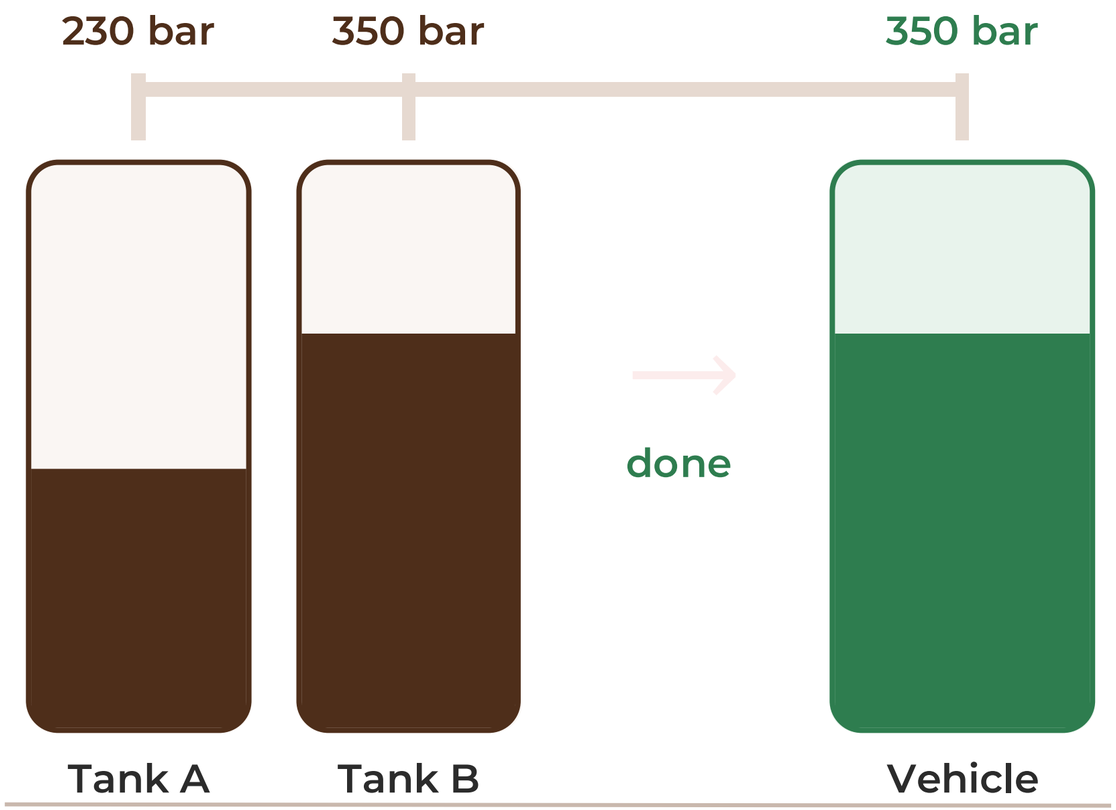


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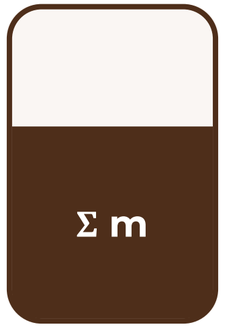
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Delivered mass = $\Delta m_A + \Delta m_B$, set by pressures and volumes, not by stored mass.

The usual premise

The station as one tank

AVAILABILITY, IN THE LITERATURE



the station as one tank

$$\Sigma m_{tanks} \geq m_{demand}$$

enough mass in store, the fill can be served

Rule-based	fill fast, in a fixed order
Optimisation-based	balance on the total mass, lumped demand

40 kg four tanks at 330 bar, request 16 kg

one tank: 40 ≥ 16, serve

cascade: no source above 350 bar, fill fails

22 kg two tanks at 400 and 500 bar, request 16 kg

one tank: less in store

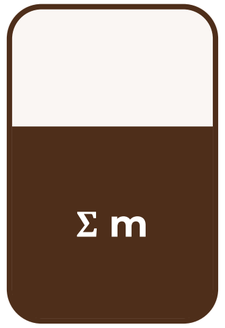
cascade: 400 → 500 bar, fill completes

Availability is a property of the stored mass AND pressure distribution. The station is not one tank.

The usual premise

More mass, and the fill fails

AVAILABILITY, IN THE LITERATURE



the station as one tank

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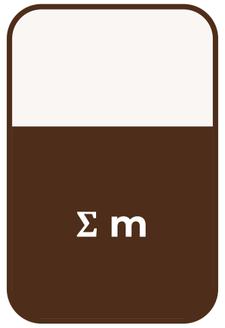
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Availability is a property of the stored mass AND pressure distribution. The station is not one tank.

The usual premise

Less mass, and the fill completes

AVAILABILITY, IN THE LITERATURE



the station as one tank

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enough mass in store, the fill can be served

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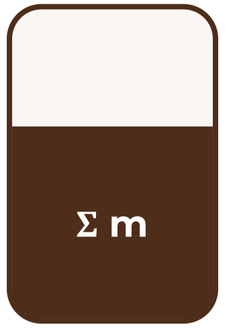
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Availability is a property of the stored mass AND pressure distribution. The station is not one tank.

The usual premise

Availability is not a mass

AVAILABILITY, IN THE LITERATURE



the station as one tank

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enough mass in store, the fill can be served

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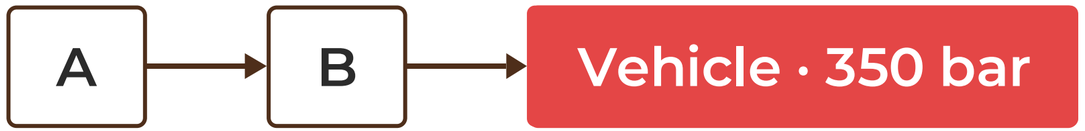
Availability is a property of the stored mass AND pressure distribution. The station is not one tank.

Refuelling patterns as a modelling strategy

An ordered sequence of sources

A pattern is an ordered sequence of sources

1



order matters:
B → A stalls at B's equilibrium

2 Step ends at **350 bar** or at **equalisation**

3 Length ≤ **2 steps**, repeats pruned → **36 candidates**

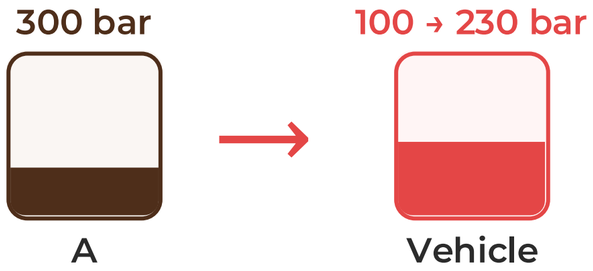
4 Available mass per pattern, **linear** in tank masses

Refuelling patterns as a modelling strategy

What one step does

1 Pattern = **ordered** sequence of sources

2 **A step opens one source and ends at the first of two events**



✓ vehicle reaches 350 bar

= P_A equalises with P_{veh}

3 Length ≤ **2 steps**, repeats pruned → **36 candidates**

4 Available mass per pattern, **linear** in tank masses

Refuelling patterns as a modelling strategy

At most two steps: 36 candidates

1 Pattern = **ordered** sequence of sources

2 Step ends at **350 bar** or at **equalisation**

3 **Length capped at 2 steps; repeats pruned**

1 step	A	**6**	A A ≡ **A** no gradient left
2 steps	A → B	**30**	A B B ≡ **A → B**
3 steps	A → B → C	0	A B A ✕ A already equalised

4 Available mass per pattern, **linear** in tank masses

Refuelling patterns as a modelling strategy

Available mass per pattern

1. Pattern = **ordered** sequence of sources
2. Step ends at **350 bar** or at **equalisation**
3. Length ≤ **2 steps**, repeats pruned → **36 candidates**
4. **Available mass of a pattern, in one algebraic step**

Pattern $T3_2$ 430 bar → T4 720 bar , vehicle at 100 bar asks 16 kg



$m^{av} = \Delta m^{(1)} + \Delta m^{(2)} = \mathbf{17.5\ kg} \geq 16$ kg → feasible

INPUTS PER CLIENT

- vehicle tank volume
- mass on board at arrival

Previous work · Applied Energy 2025

Refuelling patterns as a modelling strategy

Feasible patterns share tanks

1. Pattern = **ordered** sequence of sources
2. Step ends at **350 bar** or at **equalisation**
3. Length ≤ **2 steps**, repeats pruned → **36 candidates**
4. Available mass per pattern, **linear** in tank masses

The availability objective

Built online, from what each pattern can extract

1

INPUT

Hypothetical client

mass on arrival

hydrogen demand

2

MODEL

Available mass per pattern

closed form

linear in tank masses

3

THRESHOLD

Feasibility

available mass ≥ demand

per pattern

4

ONGOING

Availability

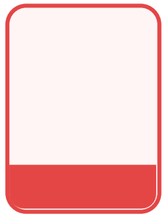
disjoint count ≥ target

The availability objective

Who must the station stay ready for?

1 A hypothetical client
the one the station must stay ready for

mass on arrival



vehicle

→ asks 16 kg

ALONG THE HORIZON



both are parameters: they can change with the time of day

Not a booked client. The objective is written for whoever might arrive.

2 MODEL
Available mass per pattern

3 THRESHOLD
Feasibility

4 ONGOING
Availability

The availability objective

What can each pattern extract for that client?

1

INPUT

Hypothetical client

2 **Available mass per pattern**

closed form, linear in the tank masses

A · 300 bar → B · 480 bar → vehicle

what this pattern can extract

m^{av} = 17.5 kg



one expression per pattern, evaluated next to the dynamics at every step. No extra states.

The solver sees what each pattern can extract, not what the tanks hold.

3

THRESHOLD

Feasibility

4

ONGOING

Availability

The availability objective

Which patterns are feasible?

1

INPUT

Hypothetical client

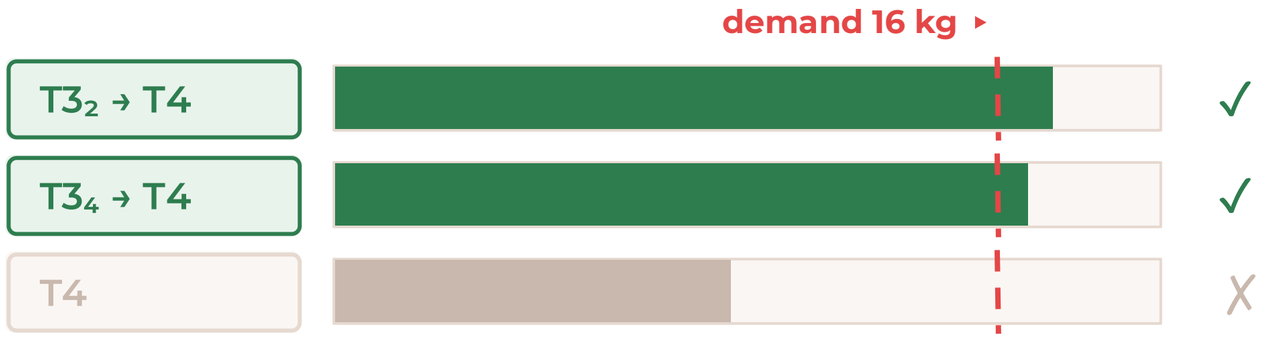
2

MODEL

Available mass per pattern

3 **Feasibility**

a threshold on the available mass, per pattern



Feasible: the pattern delivers at least the demand. The set changes with every tank pressure.

4

ONGOING

Availability

The availability objective

How many, without sharing tanks?

1

INPUT

Hypothetical client

2

MODEL

Available mass per pattern

3

THRESHOLD

Feasibility

4 **Availability**

count the disjoint feasible patterns against a target

ONGOING WORK

FEASIBLE PATTERNS

$T3_2 \rightarrow T4$ | ~~$T3_4 \rightarrow T4$~~ | $T3_1 \rightarrow T3_3$

two share T4: only one of them can count

DISJOINT		TARGET		AVAILABLE
2	≥	2	⇔	✓

The target is time-varying: raise it at rush hours, relax it at night.

The availability objective

Not a stored mass: a quantity to optimise

1

INPUT

Hypothetical client

mass on arrival

hydrogen demand

2

MODEL

Available mass per pattern

closed form

linear in tank masses

3

THRESHOLD

Feasibility

available mass ≥ demand

per pattern

4

ONGOING

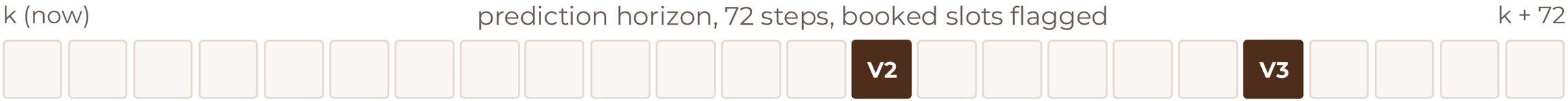
Availability

disjoint count ≥ target

Availability is a quantity the solver optimises, not directly a stored mass. All of it can be **time-variant**.

A booking enters

Three accepted bookings mark slots of the 12 h horizon



AT V1'S SLOT, K+8: 16 KG ASKED

$T3_2 \rightarrow T4$	$T3_4 \rightarrow T4$	$T4$
$T3_1 \rightarrow T3_3$	$T3_3 \rightarrow T3_2$	$T3_4 \rightarrow T3_2$

$\sum$ selected = booked flag · selected ≤ feasible

Only the selected pattern passes its per-tank masses through

TANK STATE AT K+9, THE SAMPLE AFTER V1

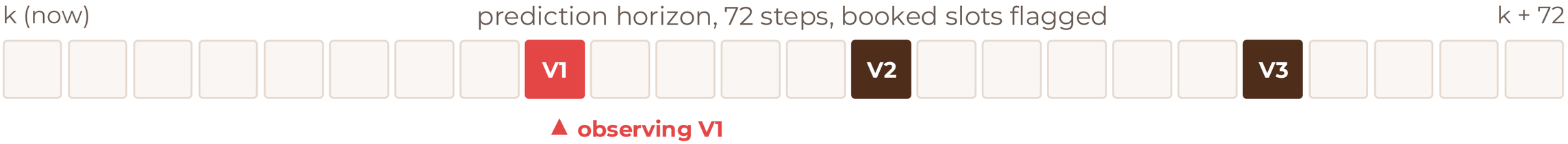
next state = state + compression − reserved



Horizon rolls: tanks moved, **selection re-optimised**, commitments kept

A booking enters

At V1's slot a pattern is required



AT V1'S SLOT, K+8: 16 KG ASKED

$T3_2 \rightarrow T4$	$T3_4 \rightarrow T4$	$T4$
$T3_1 \rightarrow T3_3$	$T3_3 \rightarrow T3_2$	$T3_4 \rightarrow T3_2$

Σ selected = booked flag but selected ≤ feasible

Only the selected pattern passes its per-tank masses through

TANK STATE AT K+9, THE SAMPLE AFTER V1

next state = state + compression − reserved



Horizon rolls: tanks moved, **selection re-optimised**, commitments kept

A booking enters

Feasible patterns for V1: its mass on board, 16 kg asked

k (now) prediction horizon, 72 steps, booked slots flagged k + 72



▲ observing V1

AT V1'S SLOT, K+8: 16 KG ASKED

$T3_2 \rightarrow T4$	$T3_4 \rightarrow T4$	$T4$
$T3_1 \rightarrow T3_3$	$T3_3 \rightarrow T3_2$	$T3_4 \rightarrow T3_2$

Σ selected = booked flag **BUT** selected ≤ feasible

Only the selected pattern passes its per-tank masses through

TANK STATE AT K+9, THE SAMPLE AFTER V1

next state = state + compression − reserved



Horizon rolls: tanks moved, **selection re-optimised**, commitments kept

A booking enters

Exactly one pattern is selected

k (now) prediction horizon, 72 steps, booked slots flagged k + 72



▲ observing V1

AT V1'S SLOT, K+8: 16 KG ASKED

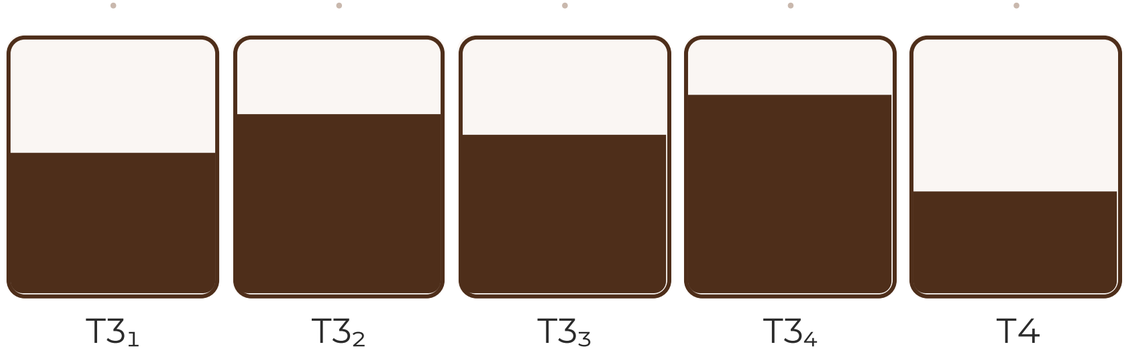
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Σ selected = booked flag **BUT** selected ≤ feasible

Only the selected pattern passes its per-tank masses through

TANK STATE AT K+9, THE SAMPLE AFTER V1

next state = state + compression − reserved



Horizon rolls: tanks moved, **selection re-optimised**, commitments kept.

A booking enters

Its per-tank masses are subtracted from the next state



AT V1'S SLOT, K+8: 16 KG ASKED

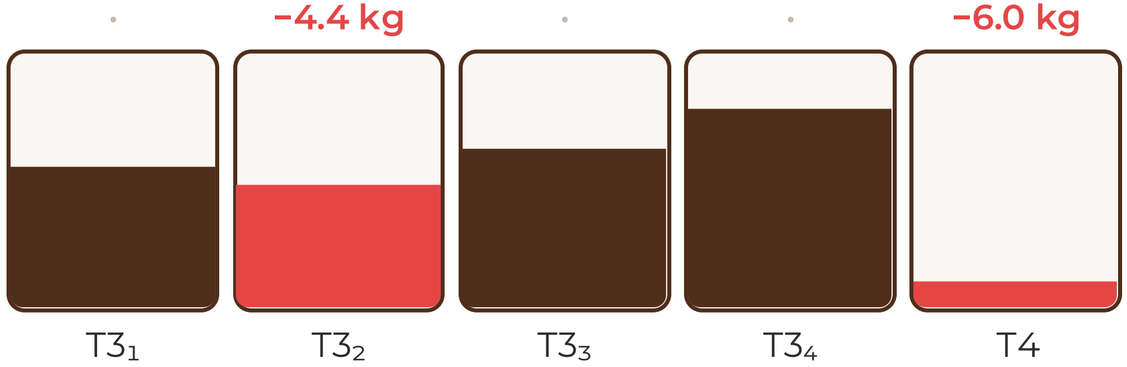
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Σ selected = booked flag **BUT** selected ≤ feasible

Only the selected pattern passes its per-tank masses through

TANK STATE AT K+9, THE SAMPLE AFTER V1

next state = state + compression − reserved



Horizon rolls: tanks moved, **selection re-optimised**, commitments kept.

A booking enters

Horizon rolls: same commitments, new selection

k + 1 (10 min later) prediction horizon, 72 steps, booked slots flagged k + 72



▲ observing past V1

AT V1'S SLOT, NOW K+7: SELECTION RE-OPTIMISED

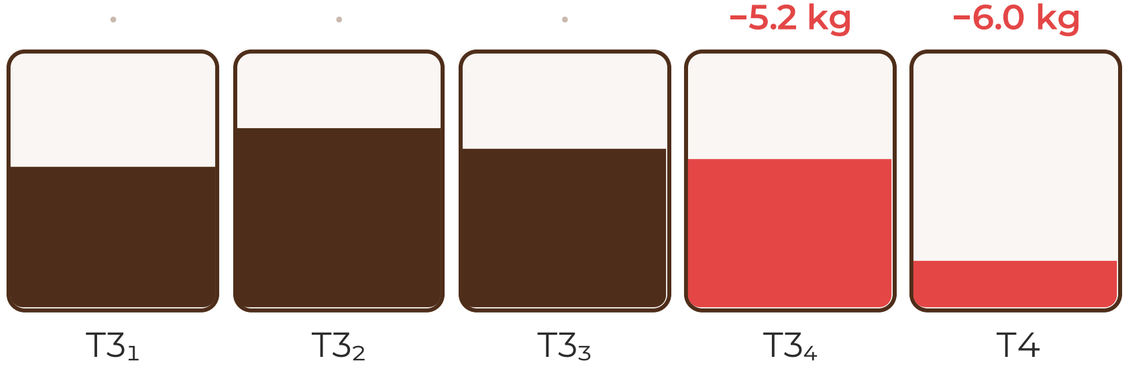
$T3_2 \rightarrow T4$	$T3_4 \rightarrow T4$	$T4$
$T3_1 \rightarrow T3_3$	$T3_3 \rightarrow T3_2$	$T3_4 \rightarrow T3_2$

Σ selected = booked flag **BUT** selected ≤ feasible

Only the selected pattern passes its per-tank masses through

TANK STATE AT K+8, THE SAMPLE AFTER V1

next state = state + compression − reserved



Horizon rolls: tanks moved, **selection re-optimised**, commitments kept.

A preliminary case

A short synthetic showcase, deterministic closed loop

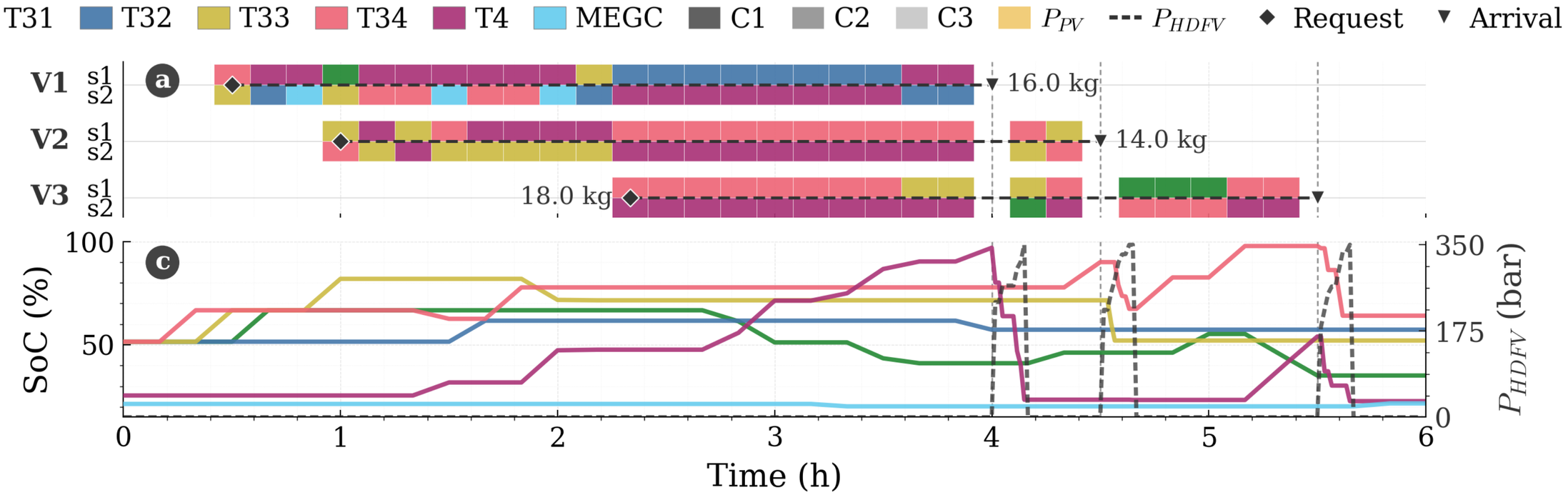
6 h winter afternoon, PV fading before the arrivals

3 bookings placed while PV is up, arrivals after dark

Client	Request	Lead	Arrival
V1	16 kg	210 min	18:00
V2	14 kg	210 min	18:30
V3	18 kg	190 min	19:30

Reserved pattern along each booking

Steps s1 and s2 of the selected pattern at every supervisory decision, from request ◆ to arrival ▼



Multiple reassignments

the solver changes the reserved pattern of each client

25 → 97 %

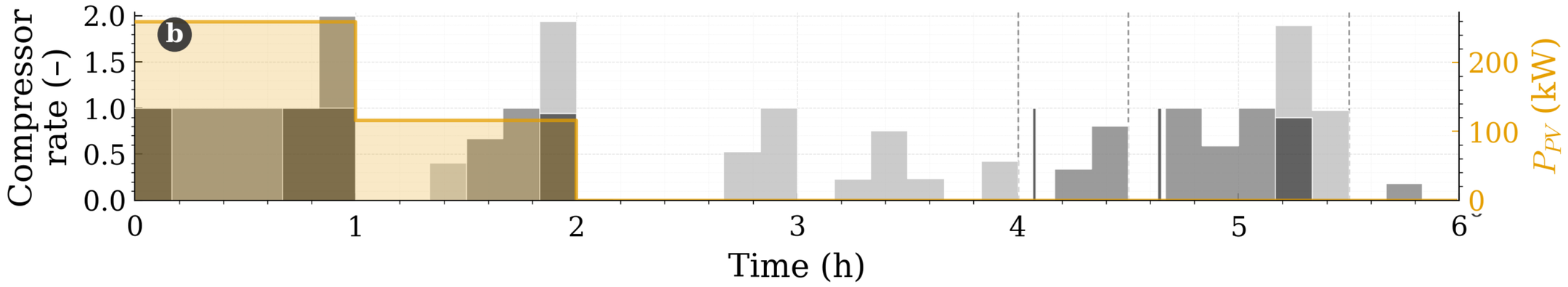
T4 raised for V1, then rebuilt only to **54 %** for V3

0

commitments broken while new bookings arrived

Compression timed to PV and arrivals

Compressor duty (C1, C2, C3) against PV power; dashed lines mark the three arrivals



Compression **front-loaded** into the last two PV hours.

Compressors **mostly off** in the dark; no eager refilling of emptied tanks.

A short burst ends **just before** each arrival, moving just the mass of the reserved pattern.

Conclusions

THIS PAPER

Bookings enter as **hard constraints**, tank by tank.

One **reserved pattern** per client, with a tunable safety margin.

Re-decided every ten minutes: **3 of 3** served on time, compression inside the PV window.

TO ANSWER THE RESEARCH QUESTION

Benchmark against rule-based operation.

More scenarios, longer runs.

Disjoint availability target effect on KPIs.

Model mismatch from real operation.

Thank you

Questions

European Fuel Cells and Hydrogen | Piero Lunghi Conference 2026 | Capri

Pol Cardona | IRI, CSIC-UPC

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